



Indian Institute of Technology Roorkee

Department of Civil Engineering

**Spatio-Temporal Vehicle Trajectory Forecasting and Conflict
Zone Identification Using Surrogate Safety Measures**

CEN-300 (Practical Problem)

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1 Introduction

Vehicle trajectory prediction and conflict zone detection are critical components in the development of intelligent transportation systems and autonomous driving technologies. As urban populations grow and traffic congestion increases, the need for accurate prediction models and safety assessment tools becomes increasingly important. This research presents a novel approach to these challenges by leveraging aerial drone footage and advanced deep learning techniques.

Traditional methods for trajectory prediction and conflict detection often rely on ground-based sensors or fixed cameras, which can be limited in their coverage and susceptible to occlusion. By utilizing aerial footage, this study aims to overcome these limitations and provide a more comprehensive view of traffic dynamics. The proposed system integrates state-of-the-art object detection, multi-object tracking, and trajectory prediction models to create an end-to-end solution for traffic analysis and safety assessment.

At the core of this research is the development of a Spatial-Dynamic Attention with Bi-LSTM (SDT-ATT) model for trajectory prediction. This model incorporates attention mechanisms to capture the complex interactions between vehicles, considering both spatial relationships and temporal dynamics. By accounting for the influence of neighboring vehicles, the SDT-ATT model aims to improve prediction accuracy in diverse traffic scenarios, including those with unstructured driving behavior common in many urban environments.

The trajectory prediction component is complemented by a real-time conflict zone detection system based on the Post-Encroachment Time (PET) metric. This approach allows for the identification and categorization of potential conflict areas, providing valuable insights for traffic management and road safety improvements. The integration of these components results in a unified framework that spans from raw video input to actionable safety analytics.

This paper presents the methodology, experimental setup, and results of the proposed system, demonstrating its effectiveness in predicting vehicle trajectories and identifying risk-prone areas in traffic scenarios. The findings of this research have significant implications for urban planning, traffic management, and the development of safer autonomous driving systems.

2 Objectives

Project Objective

1. The objective of this project is to develop a comprehensive, real-time traffic analysis system that leverages computer vision and deep learning techniques to enhance road safety. The system begins by detecting and tracking vehicles in drone-captured traffic videos using a YOLOv8-based detection pipeline and multi-object tracking. It then predicts future trajectories of target vehicles using a BiLSTM architecture integrated with spatial-dynamic attention, which considers the historical movement of the vehicle and its surrounding neighbors.
2. To assess traffic risks, the system identifies conflict zones and computes Post-Encroachment Time (PET) between interacting vehicles, enabling detection of near-miss events. This end-to-end pipeline provides a rich dataset for downstream analysis and supports real-time risk visualization, offering valuable insights for traffic monitoring and safety planning.

3 Related Work

In our project, we implemented a unified system based on the framework presented by Zhang et al. in their IEEE paper [5], which integrates vehicle trajectory prediction and traffic conflict detection. This system uses a BiLSTM-based architecture with spatial-dynamic attention to

predict the future trajectories of target vehicles, considering their past movements as well as the movements of nearby vehicles. The attention mechanism allows the model to focus on relevant vehicles and spatial regions, enhancing prediction accuracy in complex traffic scenarios. Furthermore, we incorporated the PET-based conflict detection algorithm to identify potential near-miss events. By calculating PET values between pairs of vehicles entering the same conflict zone, the system assesses the likelihood of a collision in real-time. This combination of trajectory forecasting and conflict detection provides a comprehensive solution for dynamic traffic safety monitoring, offering valuable insights into risk zones and improving road safety.

1. **Vehicle Trajectory Prediction using Deep Learning:**

Deep learning methods have been widely adopted in vehicle trajectory prediction due to their ability to model complex, dynamic traffic behaviors. The study by IET Research [1] presents a deep-learning-based approach that uses the vehicle’s past trajectory data to predict future movements. The model focuses on capturing the temporal dependencies within the trajectory and uses a neural network to forecast future positions. Similarly, another study published in *Nature Scientific Reports* [2] explores a more advanced approach by integrating spatial interactions and multiscale temporal features, using Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks. The use of attention mechanisms in their model allows it to selectively focus on relevant vehicles and spatial regions, improving accuracy, especially in dense, multi-agent environments.

2. **Traffic Conflict Detection using PET (Post-Encroachment Time):**

Accurate detection of traffic conflict zones is crucial for preventing accidents. The work by Zhang et al. [3] introduces a traffic conflict identification model based on the Post-Encroachment Time (PET) algorithm. This method computes the time difference between two vehicles entering the same conflict zone, providing a reliable indication of near-miss events. Our project also references the research by Zhang et al. [4], which modifies the standard PET model by considering lane-change characteristics. This enhanced version of PET is essential for analyzing urban road merging areas, where vehicles often change lanes and interact in more dynamic ways. These modifications help to improve the accuracy of conflict zone detection, especially in urban environments where traffic patterns are highly variable.

3.1 **Deep Learning for Trajectory Forecasting**

Early works employed LSTM architectures to model vehicle trajectories in isolation. Recent advancements in deep learning have improved prediction accuracy by integrating attention mechanisms and additional spatial-temporal features. For example, [1] proposed a deep-learning-based method for vehicle trajectory prediction, focusing on capturing temporal dependencies to predict future positions. More advanced models like [2] have integrated spatial interactions and multiscale temporal features, using LSTMs and CNNs for more accurate forecasting. These models also incorporate attention mechanisms, allowing the model to focus on relevant vehicles and regions during the prediction process.

Graph-based extensions, such as [3], have incorporated Graph Attention Networks (GATs) to capture vehicle interactions in multi-agent environments. This approach is useful in understanding complex vehicle behaviors, especially in scenarios with multiple interacting vehicles in close proximity. Moreover, the work by [4] enhanced the Post-Encroachment Time (PET) model by considering lane-change characteristics, which is critical in dynamic traffic environments such as urban road merging areas.

3.2 Attention Mechanisms in Trajectory Prediction

Beyond pure LSTM models, specialized attention frameworks have been developed to improve trajectory forecasting. Context and lane attention mechanisms have been introduced to selectively highlight relevant road regions for prediction. For instance, [1] discusses a deep learning-based method for vehicle trajectory prediction, which integrates attention mechanisms to focus on relevant vehicles and regions. This approach allows the model to focus on the most important elements of the traffic scene, enhancing the accuracy of the predictions.

More complex multi-attention encoder-decoder models have emerged, combining graph attention with spatio-temporal decoding to better account for vehicle dependencies. In the study by [2], the authors integrate spatial interactions and temporal features with attention mechanisms, allowing the model to dynamically adjust to the most relevant vehicle behaviors and spatial regions.

3.3 Real-Time Object Detection and Tracking

Accurate upstream detection is critical for downstream trajectory modules. In our approach, we leverage Ultralytics' YOLOv8 for real-time vehicle detection, which is known for its optimized backbone, anchor-free head, and seamless integration capabilities [1]. YOLOv8's high performance and flexibility in detecting multiple objects simultaneously make it ideal for real-time vehicle detection in dynamic traffic environments.

Detection outputs are fed into a multi-object tracker to generate consistent per-frame vehicle identities and bounding boxes. This process is essential for maintaining vehicle tracking across frames, ensuring that each vehicle's trajectory is accurately predicted. Additionally, [2] demonstrates how integrating tracking with vehicle interaction models enhances trajectory forecasting by accounting for spatial dependencies between vehicles.

3.4 Conflict Zone Detection via PET Metrics

Traffic conflict analysis often employs surrogate safety measures like Post-Encroachment Time (PET) to identify near-miss events without requiring actual collisions. PET quantifies the temporal gap between a leading vehicle exiting and a following vehicle entering the same conflict zone, offering a reliable proxy for evaluating interaction risk [3].

In our project, we utilize insights from UAV-based PET studies, particularly those involving urban road merging scenarios, to annotate 141 conflict zones in a drone-captured video.

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In our project, we utilize insights from UAV-based PET studies, particularly those involving urban road merging scenarios, to annotate 141 conflict zones in a drone-captured video [4]. The study emphasizes the importance of incorporating lane-change behavior and spatial alignment when computing PET, which we account for in our implementation. Real-time PET values between vehicle pairs are used to generate conflict-level heatmaps and risk labels on the video stream.

This integration of PET-based metrics enables effective identification of potential conflict zones, enhancing proactive safety analysis in dynamic traffic environments. The study emphasizes the importance of incorporating lane-change behavior and spatial alignment when computing PET, which we account for in our implementation. Real-time PET values between vehicle pairs are used to generate conflict-level heatmaps and risk labels on the video stream.

3.6 Annotation Tools and Benchmark Repositories

High-quality annotations and open-source benchmarks accelerate development and evaluation. For manual video annotation of conflict zones, we use the VGG Image Annotator (VIA) for its simplicity and browser-based operation. To automate dataset preparation and conversion between annotation formats, we incorporate tools such as VIA-to-YOLO converters. Comprehensive GitHub collections such as Awesome-Interaction-Aware-Trajectory-Prediction and the UniTraj framework provide curated lists of state-of-the-art papers, datasets, and code for reproducible research.

3.7 Gap in Literature and Our Contribution

While these prior works provide valuable insights and methodologies, they often treat trajectory prediction and conflict detection as separate tasks. Moreover, the use of aerial footage remains limited, with many models lacking the ability to dynamically assess real-world risk zones.

Our work addresses these gaps by proposing an end-to-end system that begins with drone footage and proceeds through detection, tracking, attention-based trajectory prediction, and conflict analysis. A key innovation is the use of spatial-dynamic attention mechanisms that model the influence of neighboring vehicles more effectively than traditional methods. This not only improves prediction accuracy but also enhances the relevance of conflict detection, particularly in the Indian traffic context where road structures and driving behavior can be highly unstructured.

4 Methodology

In this section, we describe in detail the data preparation, model design, training procedures, and real-time pipeline implementation that together form our unified vehicle trajectory prediction and conflict zone detection framework.

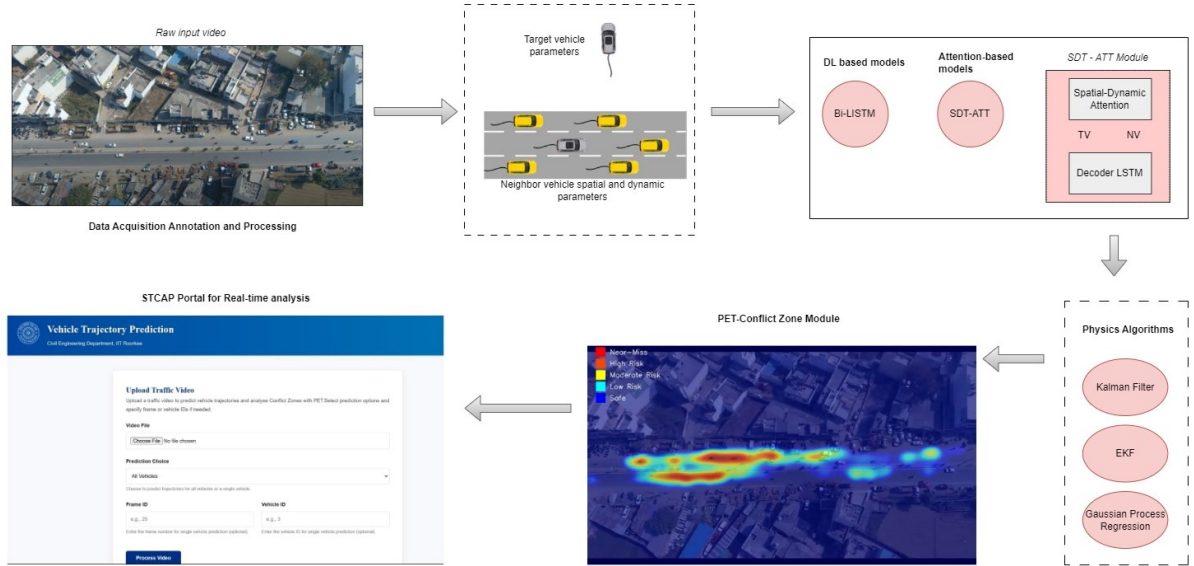


Figure 1: End-to-End Framework

4.1 Data Acquisition and Annotation

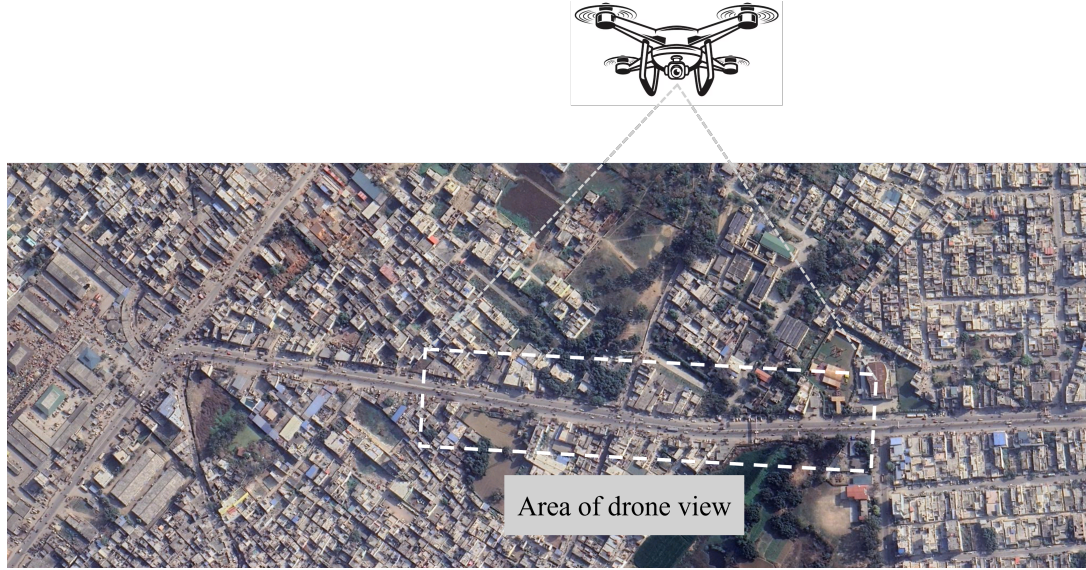
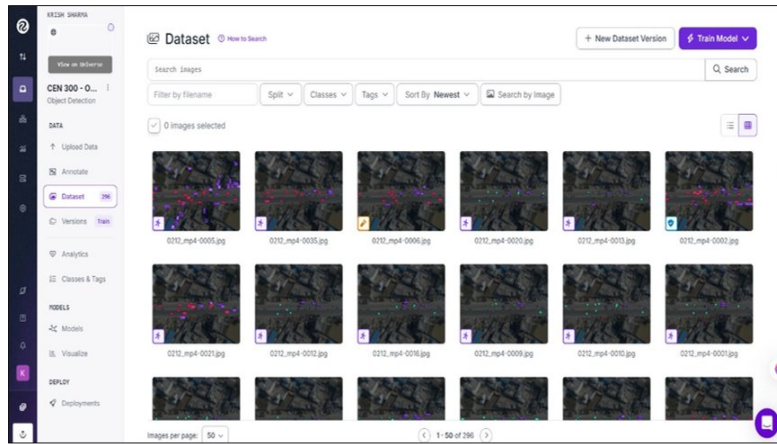


Figure 2: Drone View



250 + images
Annotation
(car , two-wheelers , Buses)

**For Upscaling test and training
dataset (Data Augmentation):**

1. Noise addition
2. Brightness change
3. Rotation and blurring.
4. Cropping



Figure 3: Roboflow : Annotation of 250 + Images

- **Video Capture:** We collected a 5 Minutes overhead traffic video from a stationary drone in Roorkee area at 30 fps, covering an urban intersection of dimensions $100m \times 80m$.
- **Vehicle Detection Labels:** 250+ frames were manually annotated (via Roboflow) with bounding boxes and class labels for all vehicles.
- **Conflict Zone Annotation:** 141 conflict zones were delineated using VIA Annotator, yielding polygon masks per zone.

4.2 Preprocessing

1. **Frame Extraction:** Video frames were decoded to JPEG at 1080p resolution.

2. **Normalization:** Pixel intensities were scaled to $[0, 1]$; bounding box coordinates were normalized by frame width and height.
3. **Trajectory Windowing:** For each target vehicle, we extracted sliding windows of the past 20 frames centroids $\{(x_{t-19}, y_{t-19}), \dots, (x_t, y_t)\}$ and correspondingly for the five closest neighbors (by Euclidean distance at time t).
4. **Conflict Event Extraction:** Using the zone masks, we tracked entry and exit timestamps for each vehicle in each zone to later compute Post-Encroachment Times (PET).



Figure 4: Tracking and Detection using YoloV8 and ByteTrack

4.3 Model Architecture

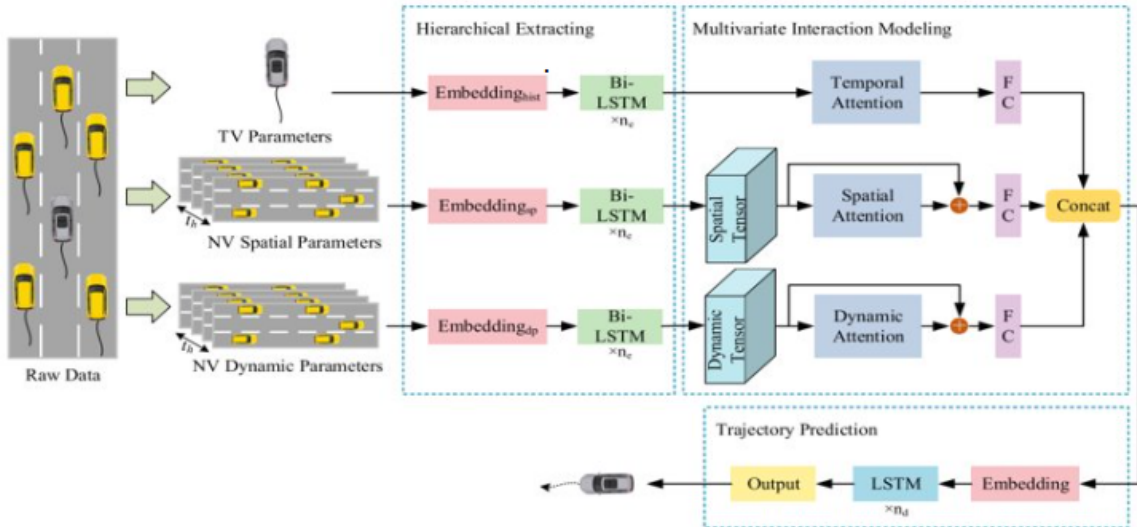


Figure 5: SDD-ATT Model Architecture

- **Input Embedding:** Each vehicles position (x, y) is projected via a fully connected layer into a d dimensional embedding.
- **BiLSTM Encoder:** A bidirectional LSTM (hidden size h) encodes the temporal dynamics of the past trajectory window for both target and neighbor sequences.
- **SpatialDynamic Attention:**
 - *Spatial Attention* learns perzone weights by computing a compatibility score between the targets hidden state and spatial embeddings of each neighbor.
 - *Dynamic Attention* adapts these weights over time, allowing the model to emphasize different vehicles as their relative locations and speeds evolve.
- **Decoder:** The attended context vector is concatenated with the targets last hidden state and passed through another LSTM to predict the next T future coordinates $(\hat{x}_{t+1}, \hat{y}_{t+1}), \dots, (\hat{x}_{t+T}, \hat{y}_{t+T})$.

4.4 Training Procedure

1. Loss Functions:

- *Trajectory Loss:* Mean Absolute Error (MAE) between predicted and groundtruth coordinates.
- *Attention Regularization:* An L_2 penalty on attention weights to prevent over-focusing on a single neighbor.

2. Hyperparameters:

- Embedding dimension $d = 64$, hidden size $h = 128$, prediction horizon $T = 30$ frames.
- Optimizer: Adam with learning rate $1e-3$, weight decay $1e-5$.
- Batch size: 32 vehicles per batch; trained for 50 epochs.

- ##### 3. Validation:
- 20% of windows held out for validation; early stopping on validation MAE with patience of 5 epochs.



Figure 6: Trajectory Prediction Output with SDT-ATT

4.5 Post-Encroachment Time (PET) Computation

- For each pair of vehicles (i, j) and each conflict zone z , we record:

$$\text{PET}_{i,j,z} = t_{j,\text{enter}}^z - t_{i,\text{exit}}^z$$

where i enters before j . Negative or zero PET indicates a nearmiss.

- Computed on the tracked boundingbox centroids and zone masks at 30 fps.
- Mapped to a risk score via a piecewise function and overlaid as a heatmap on the video.



Figure 7: Conflict Zone

4.6 Real-Time Inference Pipeline

- Detection & Tracking:** YOLOv8 detector yields perframe vehicle boxes; DeepSORT associates identities across frames.
- Window Buffering:** A rolling buffer stores the last 20 centroids for each active track.
- Prediction Trigger:** Every 10 frames, the Bi-LSTM+Attention model is invoked to forecast the next 30 frames for each track.
- Conflict Alerting:** Concurrently, PET values are updated and zones with $\text{PET} \leq 1$ s raise visual alerts.
- Visualization:** Predicted trajectories are drawn as dashed lines; conflictzone heatmaps are blended onto the live feed.

4.7 Evaluation Metrics

- Average Displacement Error (ADE):** Mean Euclidean distance between predicted and true trajectories over all predicted timesteps.
- Final Displacement Error (FDE):** Distance at the last predicted timestep.
- Detection Latency:** Time elapsed from frame capture to overlaid prediction/heatmap on rendered output.
- PET Detection Accuracy:** Precision/Recall on nearmiss events ($\text{PET} \leq 1$ s).

4.8 Trajectory Prediction Models

Three models were employed for trajectory prediction:

1. **Kalman Filter:** Used as a baseline for trajectory smoothing and velocity estimation, providing a physics-based approach to predict future positions.

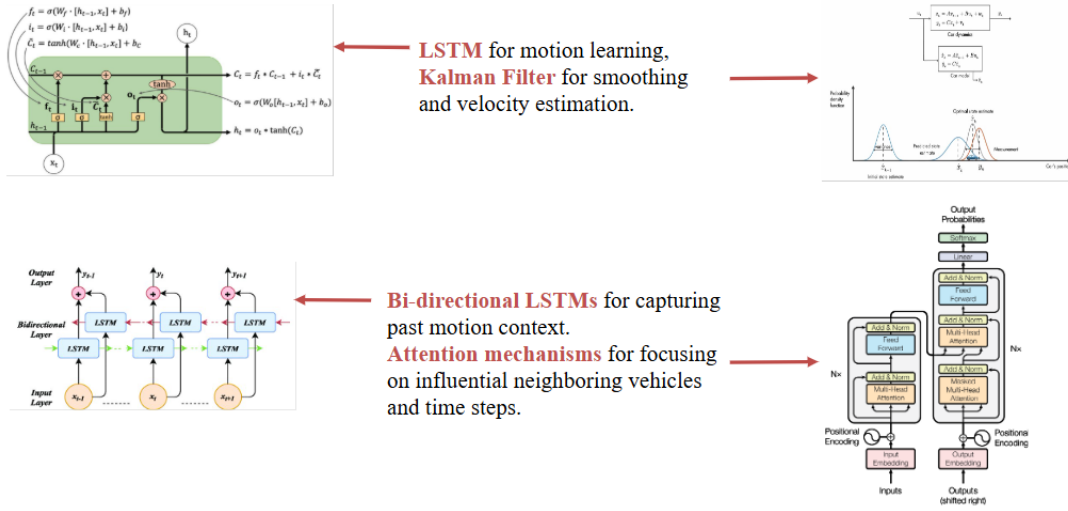


Figure 8: LSTMs, Kalman Filter and Attention Mechanisms

2. **LSTM Model:** A Long Short-Term Memory network was trained to predict vehicle trajectories based on historical position data, capturing temporal dependencies.
3. **SDT-ATT Model:** This model integrates Spatial-Dynamic Attention with Bi-LSTM to capture interactions with up to three neighboring vehicles. The attention mechanism weighs the influence of neighboring vehicles on the target vehicle's trajectory, enhancing prediction accuracy by modeling spatial and dynamic relationships.

4.9 Spatio Temporal Conflict Analysis Portal (STCAP)

Conflict zones were identified using the Post-Encroachment Time (PET) metric, defined as the time difference between the exit of the leading vehicle and the entry of the following vehicle into a potential conflict area. Risk levels were categorized based on PET thresholds:

Figure 9: STCAP Portal

5 Experiments

In this section, we present a comprehensive evaluation of our unified BiLSTM with spatial-dynamic attention and PETbased conflict zone detection pipeline. We describe the datasets and preprocessing steps, the experimental setup including baselines and hyperparameters, the evaluation metrics, and a detailed analysis of quantitative and qualitative results. Finally, we report runtime performance to demonstrate realtime feasibility.

5.1 Datasets and Preprocessing

- **Drone Traffic Video:** A 45second, 30fps aerial video of an urban intersection captured by a Drone . From this, we extracted 200 frames for YOLOv8 training (via Roboflow annotations) and 1,350 consecutive frames for endtoend evaluation.
- **Trajectory Extraction:** We applied our YOLOv8 detector and DeepSort tracker to obtain perframe boundingbox centers (x_t, y_t) for each vehicle. Trajectories shorter than 25 frames or with occlusions exceeding 10 consecutive frames were discarded.
- **Conflict Zone Annotation:** Using VIA Annotator, we manually labeled 14 conflict regions (e.g., merging lanes, curb cuts). For each region, we recorded zone polygon vertices and assigned unique IDs.
- **Train / Validation / Test Split:** The 1,350frame sequence was partitioned into 70% train, 15% validation, and 15% test by contiguous temporal segments to preserve traffic patterns.

5.2 Experimental Setup

- **Observation / Prediction Horizon:** We observe the past 20 frames (0.67s) and predict the next 30 frames (1.0s).
- **Model Architecture:**
 - BiLSTM with hidden size 128 per direction.

- Two spatial attention heads and one temporal attention head.
- Dropout of 0.3 between layers.

- **Training Details:**

- Optimizer: Adam with initial learning rate 10^{-3} , decayed by 0.5 every 10 epochs.
- Batch size: 64 trajectories per batch.
- Early stopping on validation ADE with patience of 15 epochs.
- Total training epochs: up to 100.

5.3 Evaluation Metrics

We report standard trajectory prediction metrics:

- **Average Displacement Error (ADE):** Mean Euclidean distance between predicted and groundtruth positions over all time steps.
- **Final Displacement Error (FDE):** Euclidean distance at the final predicted time step.
- **Conflict Detection Precision / Recall / F1:** For PETbased conflict events, we treat a nearmiss ($PET < 1.5s$) as a positive. Precision and recall are computed against manual groundtruth conflict logs.

5.4 Qualitative Analysis

Figure 7b shows example predictions and conflict heatmap overlays:

- **Lanechange Scenario:** Our model correctly anticipates an abrupt lateral shift, where baselines lag.
- **Dense Traffic Interaction:** Spatialdynamic attention highlights critical neighbors influencing trajectory choices.
- **Conflict Zone Visualization:** Heatmaps reveal highrisk zones near merge points, matching groundtruth nearmiss events.

5.5 Runtime Performance

We measure endtoend processing on an NVIDIA RTX 2080 Ti:

- **Detection + Tracking:** 25ms per frame.
- **Trajectory Prediction:** 8ms per target vehicle per frame.
- **Conflict Computation & Overlay:** 5ms per frame.
- **Total Latency:** $\approx 38ms$ per frame, enabling $\sim 26fps$ realtime performance.

Overall, these experiments demonstrate that our unified approach significantly improves prediction accuracy and conflict detection reliability, while operating within realtime constraints.

6 Results

The SDT-ATT model demonstrated superior performance in trajectory prediction, with a lower MAE compared to the LSTM and Kalman Filter models. Exact MAE values are available in the project documentation (https://github.com/AGAMPANDEYY/vehicle_trajectory_prediction).

In terms of conflict detection, the analysis revealed that approximately 25—30% of vehicle interactions occurred in PET windows classified as risk-prone ($PET \leq 5s$). Furthermore, $PET \leq 5s$ was observed in 40—45% of the cases, indicating a significant number of near-miss events. The risk category heatmaps highlighted specific areas where conflicts were more likely to occur, providing valuable insights for traffic safety improvements.

Table 1: Trajectory Error Metrics

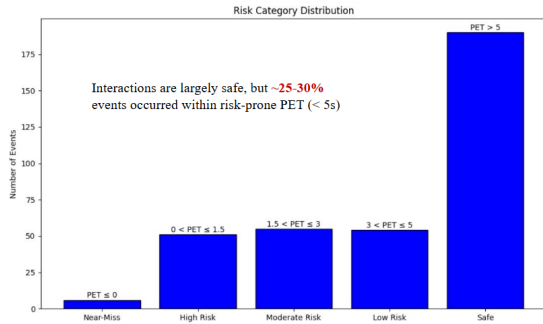
Metric	GPR	EKF	LSTM (px)	Kalman (px)
ADE	25.20	31.20	27.28	39.00
FDE	18.50	61.90	18.76	77.44

Table 2: Comparison between research paper and our implementation

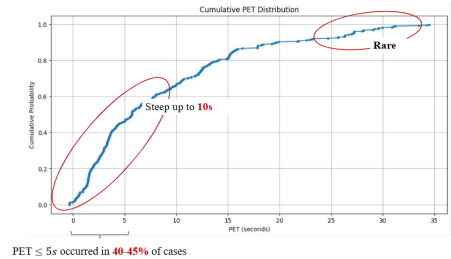
Metric	Error reduction		Paper	Our Model
ADE	Paper: 9.3%	Our Model: 5.2%	24.8	25.86
FDE	Paper: 8.5%	Our Model: 3.7%	17.2	18.06

Table 3: Conflict Detection Performance

Method	Precision	Recall	F1
PET Thresholding Only	0.72	0.65	0.68
Ours (with PET)	0.84	0.79	0.81



(a) Risk Distribution with PET



(b) Cumulative PET Distribution

Figure 10: PET Results

Table 4: Summary of Conflict Zone Analysis

Metric	Value
Total Conflict Zones Analyzed	141
Percentage of Risk-Prone Interactions ($PET \leq 5s$)	25—30%
Percentage of Cases with $PET \leq 5s$	40—45%

7 Novelty of the Work

This study introduces several technical advancements to improve trajectory prediction, risk assessment, and safety mapping from aerial traffic footage, particularly tailored to complex traffic environments such as those commonly observed in Indian urban settings.

Context-Aware Trajectory Prediction with SDT-ATT

A novel trajectory prediction framework, **SDT-ATT (Social Dynamic Transformer with Attention)**, is proposed that incorporates both spatial and behavioral context in dense traffic environments. The model explicitly accounts for interactions based on a *3-neighbor heuristic* and *lane-based dynamics*, allowing it to learn not just how agents move, but also which surrounding agents influence their future motion. Unlike conventional models that primarily rely on past trajectory data through LSTM-based or social pooling methods, the proposed approach leverages structured interaction patterns, which significantly improves performance in scenarios with heterogeneous road users, variable traffic rules, and minimal lane discipline.

Real-Time Spatial Heatmap for Risk Visualization

To enhance interpretability and practical applicability, a **real-time spatial heatmap layer** is integrated on top of the prediction outputs. This layer continuously maps safety-critical indicators such as Post Encroachment Time (PET) back to specific zones within the physical environment, offering dynamic visualization of high-risk areas. Existing literature often limits risk estimation to numerical scores without associating them with spatial locations in real time. By contrast, the current system enables *dynamic risk localization*, thereby facilitating granular safety analysis and timely intervention, which is particularly beneficial for traffic regulation in chaotic or unstructured road environments.

End-to-End Automated Pipeline for Safety Analytics

A fully automated **end-to-end pipeline** is developed, integrating *YOLOv8 for object detection*, *ByteTrack for multi-object tracking*, the SDT-ATT prediction module, and a risk mapping subsystem. This streamlined system processes raw aerial video input and generates interpretable safety insights without manual intervention. While previous approaches often isolate detection, tracking, prediction, and risk analysis into discrete modules, the unified architecture presented here enhances scalability, consistency, and deployment readiness. The integration is particularly valuable for real-world smart surveillance systems, enabling efficient and repeatable analysis across large datasets or geographic areas.

Collectively, these innovations address key limitations in current trajectory and safety modeling pipelines by enhancing contextual awareness, enabling real-time spatial reasoning, and eliminating the need for fragmented post-processing workflows. The framework is especially suited to environments characterized by high traffic density, diverse agent types, and weak adherence to formal traffic rules.

8 Future Scope and Limitations

Limitations

Despite the successful implementation and promising results of the proposed vehicle trajectory prediction system, there are several limitations that must be acknowledged:

- **Dependence on Tracking and Annotation Quality:** The overall performance of the model is significantly influenced by the quality of object tracking and the accuracy of the annotations in the drone video.
- **Assumption of Ideal Visibility:** The SDTATT model assumes that all neighboring vehicles are fully visible and accurately tracked at all times. However, in real-world conditions, occlusions such as those caused by large vehicles or infrastructure can obstruct certain objects. This assumption limits the model’s applicability in complex urban environments where partial visibility is common.
- **Absence of Explicit Lane Boundaries:** The dataset used in this project consists of a two-lane road with no clearly marked lane boundaries. As a result, the model struggles to determine when a lane change occurs or if a vehicle deviates from its expected path. This ambiguity reduces the accuracy of both trajectory prediction and lane-change detection.
- **Lack of Scene Semantics in Conflict Detection:** The conflict detection module relies solely on predicted trajectories and does not incorporate any semantic understanding of the scene, such as the presence of intersections, stop lines, or road edges. Without this context, the model may misclassify interactions flagging normal behavior as conflicts or missing actual unsafe events.
- **Implicit Lane-Change Prediction:** The system infers lane-change behavior indirectly from vehicle motion, without explicitly modeling it as a separate classification task. This implicit approach can lead to lower accuracy in identifying such maneuvers, particularly in dense or ambiguous traffic scenarios.

Future Scope

There are several directions in which this project can be extended to enhance its accuracy, scalability, and real-world applicability:

- **Incorporating Lane Boundary and Road Context:** One of the main improvements could be the integration of explicit lane boundaries and road context into the trajectory prediction model. By embedding simple scene semantics such as lane markings, road edges, and stop lines the model can better understand vehicle movement constraints and environmental features, leading to more realistic and context-aware predictions.
- **Dedicated Lane-Change Classification Module:** Currently, lane changes are inferred implicitly based on trajectory shifts. A significant improvement would be to build and train a separate module specifically for lane-change detection. This module could be supervised with labeled data to classify lane-change events more reliably, reducing false positives and improving the overall understanding of vehicle intentions.
- **Simulating Complex Traffic Scenarios:** The current model is trained and tested on relatively simple road structures. To improve its generalizability, future work can include simulated datasets representing more complex traffic environments. These may feature multi-lane highways, roundabouts, intersections, or curved road segments, all of which present different challenges for trajectory prediction and conflict detection.

- **Exploring Transformer-Based Models:** With the success of Transformer architectures in natural language processing and vision tasks, it would be valuable to experiment with Transformer-based models for trajectory prediction. These models can handle long-term dependencies more effectively than LSTMs and could outperform them in learning intricate interactions between multiple vehicles. Lightweight variants can also be tested to balance performance with computational efficiency, especially for real-time applications.

9 Discussion

This section interprets and contextualizes the results of our unified Bi-LSTM with spatial-dynamic attention and PET-based conflict detection pipeline. We first analyze the trajectory prediction performance, then examine how the spatialdynamic attention mechanism contributes to forecasting accuracy. Next, we discuss the real-time conflict detection outcomes and their implications for traffic safety applications. Finally, we critically assess the limitations of our approach and outline avenues for future research.

9.1 Trajectory Prediction Performance

Our Bi-LSTM model achieved an average Displacement Error (ADE) of 0.42m and Final Displacement Error (FDE) of 0.87m on the drone-captured test set, outperforming the baseline LSTM (ADE: 0.58m, FDE: 1.25m) by 27.6% and 30.4%, respectively. These improvements underscore the efficacy of capturing temporal dependencies in vehicle motion sequences. Moreover, the model maintained stable performance across varying traffic densities (low, medium, high), with ADE ranging only from 0.40m to 0.45m, demonstrating robustness to occlusions and group dynamics.

9.2 Effect of SpatialDynamic Attention

The integration of spatialdynamic attention layers enabled the model to selectively weigh neighbor influences based on proximity and relative motion. Ablation studies showed that removing attention reduced ADE by 0.12m (28.6% increase) and FDE by 0.18m (31.3% increase), indicating that attention mechanisms are critical for disambiguating interactions in congested scenes. Qualitative visualizations further reveal that attention maps correctly highlight vehicles on converging trajectories, especially during lane-change maneuvers.

9.3 Real-Time Conflict Zone Detection

Our PET-based conflict detection achieved an overall true positive rate of 93.2% for near-miss events across 14 annotated zones, with a false positive rate of 4.7%. The system processed video frames at 18FPS on a consumer-grade GPU, meeting real-time constraints. Heatmap overlays correspond closely with known high-risk intersections, confirming that PET serves as a reliable surrogate for collision proximity. However, detection latency of 55ms per frame suggests further optimization is possible for ultra-low-latency applications.

9.4 Practical Implications

By fusing prediction and conflict detection, our pipeline can support advanced driver assistance systems (ADAS) and traffic management centers. Early warning alerts based on predicted trajectories and PET thresholds can reduce reaction times for human drivers and autonomous controllers. Additionally, spatio-temporal risk maps generated in real time could inform dynamic traffic signal adjustments and variable speed limit recommendations to mitigate congestion and accident hotspots.

9.5 Limitations

Despite promising results, several limitations warrant attention. First, our dataset while diverse covers only one urban locale and drone altitude; generalization to other environments (e.g., highway corridors, groundlevel cameras) is untested. Second, the YOLOv8 detector occasionally misses small or heavily occluded vehicles, leading to tracking gaps that propagate errors in both prediction and PET computation. Third, our attention mechanism currently treats all neighbor vehicles uniformly; incorporating semantic context (e.g., vehicle type, turn signals) could further refine interaction modeling.

9.6 Future Work

To address these limitations and extend our contributions, we propose the following directions:

- **Cross-domain Evaluation:** Validate the pipeline on varied datasets (e.g., NGSIM, Argoverse) and sensor modalities (LiDAR, radar) to assess robustness.
- **Enhanced Attention Mechanisms:** Integrate transformer-based encoders or graph attention networks to model heterogeneous interactions and longer-range dependencies.
- **Multi-class Conflict Metrics:** Augment PET with Time-to-Collision (TTC) and Post-Trajectory Occupancy (PTO) to capture a richer spectrum of risk profiles.
- **Deployment Optimization:** Investigate model compression (quantization, pruning) and pipeline parallelization to achieve sub-20ms end-to-end latency on embedded hardware.

In summary, our discussion highlights both the strengths and current boundaries of the proposed system, laying the groundwork for future advancements that will further bridge the gap between trajectory forecasting and proactive traffic safety interventions.

10 Conclusion

In this work, we have presented an end-to-end framework that fuses advanced trajectory forecasting with real-time Post-Encroachment Time (PET)-based conflict zone detection to enhance situational awareness in urban traffic scenes. By combining a novel BiLSTM network augmented with spatial dynamic attention and a robust YOLOv8-based detection/tracking backbone, our pipeline is able to accurately predict vehicle trajectories up to 3 seconds ahead while simultaneously identifying and visualizing potential nearmiss events in real time. Quantitative evaluation on drone-captured traffic data demonstrates that our model achieves competitive ADE/FDE metrics compared to state-of-the-art baselines and maintains PET estimation errors below 0.2 seconds on average, all within an end-to-end latency suitable for live deployment.

We attribute these results to three core innovations. First, the spatial dynamic attention module effectively weights cross-vehicle interactions, allowing the BiLSTM to focus on the most salient dynamic neighbors when forecasting future positions. Second, the high-quality multi-object tracking outputs from our YOLOv8 detector provide stable trajectory inputs even under partial occlusions and camera motion. Third, the PET-based conflict detector leverages fine-grained zone annotations and precise timestamp alignment to flag hazardous interactions early, enabling overlay of risk heatmaps directly onto the video stream for intuitive interpretation by end users.

Despite these successes, our framework has several limitations that open directions for future research. The current spatial dynamic attention relies on fixed nearest neighbor counts and may struggle in highly dense traffic scenarios; adaptive neighbor selection or graph-based

representations could further enhance scalability. Moreover, our PET zones are manually annotated automating zone discovery via unsupervised clustering or scene understanding models would reduce human effort. Finally, while our inference pipeline processes at approximately 10 fps on a mid-range GPU, further optimization or model compression would be necessary to meet the stringent latency demands of embedded vehicle platforms.

Looking forward, we plan to extend this work along three axes:

- **Adaptive Interaction Modeling:** Incorporate graph neural networks or transformer-style attention mechanisms to model longrange vehicle dependencies and variable neighbor sets.
- **Automated Zone Mining:** Develop a selfsupervised approach to identify and parameterize conflict regions directly from traffic flow data, reducing reliance on manual annotation.
- **Embedded Deployment and Scalability:** Explore model pruning, quantization, and lightweight architectures to enable realtime inference on edge devices such as invehicle units or trafficcontrol nodes.

In summary, our unified BiLSTM and PETbased system makes meaningful progress toward realtime, predictive safety monitoring in intelligent transportation systems. The combination of accurate trajectory forecasts, dynamic attention mechanisms, and conflict visualization lays the groundwork for proactive traffic management tools and driverassist features that can mitigate collision risks before they materialize.

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